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Potash Fertilizer
From Underground Salts to Crop Nutrients

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Explanation

Potash fertilizer supplies potassium, a nutrient plants need for growth and water control. Potash is a group of potassium salts; this project focuses on potassium chloride, KCl (Natural Resources Canada [NRCan], 2026; Kaiser & Rosen, 2018). Canada produces potash mainly in Saskatchewan. One important step in its industrial history was Kalium's solution mine at Belle Plaine in 1964 (Mosaic, n.d.).

Solution mining uses water to bring salts out of underground rock. Warm brine, meaning salty water, is pumped into the deposit. It dissolves potassium chloride and sodium chloride, NaCl, and the salt solution is pumped to the surface. Cooling helps KCl form solid crystals that can be collected (Saskatchewan Mining Association [SMA], n.d.). Drying and sizing prepare the product for sale as fertilizer (Nutrien, n.d.). Wells, pumps, cooling equipment and dryers are important parts of this route.

The chemistry depends on solubility, meaning how much of a substance can dissolve at a given temperature. KCl becomes less soluble as the brine cools, allowing crystals to form (SMA, n.d.). Dissolving can be shown as $KCl(s) \rightarrow K^{+}(aq) + Cl^{-}(aq)$, and crystallizing reverses that change. Here, s means solid, aq means dissolved in water, and an ion is a charged particle. These equations describe separating and recovering an existing salt, rather than making a new compound (Flowers et al., 2019). Plants can take up the potassium dissolved in soil water (Kaiser & Rosen, 2018).

Mines must be near suitable deposits. Water and energy support pumping and processing, while railways and ports connect producers with distant buyers (SMA, n.d.; Nutrien, n.d.). Canada exports potash around the world, making transport and access to customers important to the industry (NRCan, 2026). Skilled workers operate and maintain the equipment, providing jobs in mining communities (Nutrien, 2026). Chemists test product quality and help improve separation methods, supporting reliable fertilizer production (SMA, n.d.).

Potash helps farmers replace potassium when soil supplies are too low, but applying too much fertilizer near seeds can cause salt damage. Soil testing helps farmers decide how much is needed (Kaiser & Rosen, 2018). Mining also leaves salty waste. Where waste brine is injected underground, Saskatchewan requires well design, monitoring and testing intended to keep it away from groundwater (Saskatchewan Ministry of Energy and Resources, 2023).

Reusing brine can reduce waste and the demand for fresh water (SMA, n.d.). The infographic shows solution mining alongside mines that cut solid ore and process it at the surface (NRCan, 2026; Nutrien, n.d.).

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Potash process and research credits
Samuel Hopkins is credited for his 1790 patent for making potash and pearl ash (United States Patent and Trademark Office, n.d.). This early process should be distinguished from modern KCl mining. Kalium's team is credited for building the Belle Plaine solution mine in 1964 (Mosaic, n.d.). The SMA lesson credits Murray Schultz for the teaching activity and Larry Bogdan for the earlier lesson plan (SMA, n.d.). The cited sources do not identify one inventor of the whole modern process.

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